

NIFTY 100

ANALYTICS DASHBOARD

Professional Analyst Guide

FUNDAMENTAL ANALYSIS • VALUATION • SCREENING • PEER INTELLIGENCE

CASH-FLOW ANALYTICS • NLP SIGNALS • CLUSTERING • API

PROJECT PROFILE	TECHNOLOGY STACK
Internship Capstone 92 Nifty 100 companies represented End-to-end analytics workflow	Python • Pandas • NumPy • SQLite Plotly • Streamlit • FastAPI pytest • Excel • CSV • PDF

A structured operating manual for analysts, reviewers, developers and internship evaluators.

Grand Edition • 2026

Document Control & Executive Snapshot

Item	Details
Document	Analyst Guide — Nifty 100 Analytics Dashboard
Purpose	Explain the platform, analytical workflow, outputs, QA controls and analyst usage.
Primary audience	Financial analysts, data analysts, students, project reviewers and developers.
Data universe	92 companies represented in the project database.
Core database	db/nifty100.db
Dashboard	Streamlit-based interactive analytics interface.
API	FastAPI service for selected analytical endpoints.
QA artifact	reports/pytest_report.html

Executive snapshot

The Nifty 100 Analytics Dashboard is an integrated analytics platform that transforms raw financial datasets into structured company-level intelligence. The project combines data engineering, database management, financial-ratio computation, screening, peer comparison, valuation analytics, cash-flow intelligence, NLP-based signal extraction and unsupervised clustering.

The platform is designed around a simple principle: **validate the data first, analyze it second, and interpret results with context**. This guide documents that workflow and provides a repeatable reference for using the project.

Key project outputs include screener results, peer comparisons, valuation summaries, valuation flags, cluster assignments, NLP-derived pros/cons signals, automated QA reports and this analyst guide.

Guide at a Glance

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Recommended reading path

New users should read Sections 01–04 first, then use Sections 05–11 as the analytical reference. Developers and reviewers should additionally read Sections 12–15 for output structure, testing and governance.

01 — Platform Overview

The platform brings multiple financial-analysis workflows into one controlled environment. Instead of manually joining separate spreadsheets for company fundamentals, ratios, screening and peer analysis, the system creates reusable analytical layers.

What the platform does

- Loads and standardizes source financial datasets.
- Stores structured entities and financial records in SQLite.
- Calculates financial ratios and derived KPIs.
- Runs multi-condition screening and reusable presets.
- Creates peer groups and relative comparison metrics.
- Produces valuation summaries and analytical flags.
- Derives cash-flow intelligence indicators.
- Extracts structured pros/cons signals from analysis text.
- Groups companies using KMeans clustering.
- Serves selected analytics through a FastAPI layer.
- Produces exportable CSV/XLSX/PDF/HTML artifacts.

Analytical philosophy

The dashboard is a decision-support system rather than a replacement for analyst judgment. Automated outputs are intended to reduce repetitive work, improve consistency and make patterns easier to investigate.

Important distinction: a screen, flag, cluster or NLP signal is an analytical output. It is not, by itself, a conclusion about the future performance of a company.

02 — Architecture & Data Flow

The project follows a layered architecture. Each stage has a distinct responsibility, making the system easier to debug and test.

Layer	Role	Representative location
Raw data	Original source files	data/raw
Processing	Cleaning, transformation and ETL	src/ + data/processed
Database	Persistent structured storage	db/nifty100.db
Analytics	Ratios, screening, peers, valuation, clustering	src/analytics
Presentation	Interactive dashboard	src/dashboard
API	Programmatic access to selected results	src/api
QA	Regression and validation	tests/ + reports
Documentation	User and project documentation	docs/

End-to-end flow

Source files → ETL → Cleaned datasets → SQLite → Analytics engines → Dashboard/API → Exports → QA

This flow separates data preparation from presentation. A dashboard change should not require rewriting the database, and a ratio calculation change should be testable without manually inspecting every screen.

Why the separation matters

- Improves reproducibility.
- Reduces duplicated calculations.
- Allows targeted testing.
- Makes data-quality exceptions visible.
- Supports future expansion of the company universe.

03 — Data Foundation & Database

The database is the central analytical store for the project. It allows the dashboard and analytics modules to use consistent records instead of repeatedly reading raw spreadsheets.

Core database areas

Table / area	Analytical purpose
companies	Company identity and master information
sectors	Sector classification and mapping
analysis	Company analysis text/content
financial_ratios	Calculated ratio records
balancesheet	Balance-sheet information
profitandloss	Profit-and-loss information
cashflow	Cash-flow information
market_cap	Market capitalization information
stock_prices	Price-related records
peer_groups	Peer relationships
peer_percentiles	Relative peer statistics
documents	Supporting document metadata
prosandcons	Structured analytical signals

Data-quality checkpoint

The project validation cycle identified **92 companies** in the database and **90 companies with ratio coverage**. ATGL and SBIN were identified as ratio-data exceptions. These exceptions are useful QA findings because they reveal where source completeness affects downstream analytics.

Do not fabricate missing financial values merely to make a dashboard complete. A visible exception is preferable to an unsupported number.

04 — Company Overview & Dashboard Navigation

The dashboard should be approached as an analytical workspace. The analyst begins with the company or universe, verifies the context, then progressively moves into ratios, screens, peers, valuation and deeper intelligence.

Recommended navigation sequence

Step	Action	Question answered
1	Confirm universe and period	What data am I analyzing?
2	Review company profile	Who/what is the company?
3	Inspect core ratios	How is the company performing?
4	Run a screen	Which companies meet selected conditions?
5	Compare peers	How does the company compare?
6	Inspect valuation	How do valuation indicators look?
7	Review cash flow	How is cash being generated/used?
8	Review NLP signals	What textual themes were extracted?
9	Review cluster	Which analytical profile is similar?
10	Export & document	What evidence should be retained?

Dashboard hygiene

- Always check active filters before interpreting a chart.
- Record the selected period when exporting results.
- Investigate unexpected outliers instead of ignoring them.
- Use detailed tables to validate headline KPI cards.

05 — Financial Ratio Intelligence

The ratio engine converts financial statement data into standardized analytical measures. During the project validation cycle, the ratio dataset contained **1,160 rows**.

Ratio families

Family	Examples	Typical analytical question
Profitability	ROE, ROA, margins	How effectively is the business generating profit?
Liquidity	Current ratio	Can near-term obligations be covered?
Leverage	Debt-to-equity, ICR	How much financial risk is associated with debt?
Efficiency	Asset / working-capital metrics	How efficiently are resources used?
Valuation	Market-based multiples	How is the market valuing the business?

Interpretation framework

Use a three-layer interpretation model:

- **Absolute:** inspect the company's own value.
- **Historical:** compare the value across available periods.
- **Relative:** compare with appropriate peers or sector context.

A single ratio rarely explains an entire business. For example, leverage should be interpreted together with interest coverage, profitability and cash generation.

Edge cases

- Null values should remain distinguishable from zero.
- Zero denominators must be handled explicitly.
- Negative earnings can make common valuation multiples difficult to interpret.
- Extreme values should be investigated for source or calculation issues.

06 — Screener & Analytical Presets

The screener transforms analytical conditions into a repeatable shortlist. It is one of the platform's most practical tools because it allows analysts to move from a broad universe to a focused investigation set.

Implemented project presets

Preset	Observed output
Quality Compounder	23 companies
Value Pick	1 company
Growth Accelerator	18 companies
Dividend Champion	30 companies
Debt Free Blue Chip	18 companies
Turnaround Watch	33 companies

How to use presets professionally

- Treat the preset as a starting hypothesis, not a final conclusion.
- Inspect the individual filter conditions.
- Review the underlying ratios for every shortlisted company.
- Check sector concentration in the resulting list.
- Compare selected companies with peers.
- Export the exact screen configuration and result when documenting analysis.

Preset names describe implemented analytical filters. They should not be interpreted as guarantees, ratings or investment recommendations.

07 — Peer Comparison Engine

Peer comparison provides context by placing a company alongside comparable companies. The project includes peer groups and percentile-oriented analysis.

Peer-review workflow

- Identify the company's business and sector.
- Select a meaningful peer set.
- Compare profitability and efficiency.
- Compare leverage and financial coverage.
- Compare valuation indicators.
- Review percentile positioning.
- Investigate differences using underlying statements.

Analyst principle

A company should be compared with companies that have reasonably comparable business models, financial structures and operating environments. A technically correct comparison can still be analytically weak if the peer set is inappropriate.

Output artifact

output/peer_comparison.xlsx contains the project's generated peer-comparison output and can be retained as evidence for internship analysis.

08 — Valuation Analytics

The valuation module consolidates market and financial information into a structured valuation view. It also generates flags that direct attention toward unusual or noteworthy observations.

Primary outputs

Output	Purpose
valuation_summary.xlsx	Company-level valuation summary
valuation_flags.csv	Structured valuation observations
Dashboard valuation page	Interactive exploration

Valuation review checklist

- Confirm the period of the financial denominator.
- Check whether earnings or book values are positive and meaningful.
- Review valuation metrics alongside growth and profitability.
- Compare against relevant peers.
- Investigate extreme multiples before drawing conclusions.
- Document any exclusions or special handling.

Valuation is context-dependent. A low multiple can arise from many different business circumstances, and a high multiple can reflect expectations that may or may not materialize. The dashboard therefore presents analytical evidence rather than a single automated verdict.

09 — Cash-Flow Intelligence

Cash-flow analytics add a second perspective to accounting-based profitability. The goal is to help analysts understand whether reported business performance is accompanied by corresponding cash-generation patterns.

Core areas

- Operating cash flow.
- Investing cash flow.
- Financing cash flow.
- Derived cash-flow KPIs.
- Relationship between earnings and cash generation.

Practical review

Begin with operating cash flow, then understand the major investing and financing movements. A strong business can have temporarily negative free cash flow because of investment, while weak cash conversion can also indicate areas requiring deeper investigation.

Project implementation

The project includes a dedicated `cashflow_kpis.py` module and generated cash-flow intelligence as part of the broader analytics pipeline.

10 — NLP Pros/Cons Signals

The project adds a lightweight natural-language analytics layer to structured financial analysis. Text from analysis.xlsx is parsed into a structured representation and converted into confidence-scored signals.

Observed project output

Metric	Observed value
Parsed source	analysis.xlsx
Parsed output	analysis_parsed.csv
Generated output	pros_cons_generated.csv
Total signals	874
Pros	562
Cons	312
Minimum confidence	65

Responsible use

- Use NLP output as an investigation aid.
- Check the original text when a signal matters.
- Consider confidence alongside context.
- Combine textual evidence with financial evidence.
- Do not treat automated sentiment or extraction as ground truth.

NLP can accelerate reading and organization, but the final interpretation remains dependent on source quality, wording, context and the underlying financial evidence.

11 — Clustering & Company Profiles

KMeans clustering groups companies according to selected analytical characteristics. The project generated five descriptive profiles across the 92-company universe.

Cluster	Companies
Value Cyclical	45
Emerging Growth	19
High-Quality Compounds	15
Defensive Dividend Payers	11
Distressed or Turnaround	2

How to read a cluster

- Inspect the feature profile behind the cluster.
- Compare multiple companies within the same cluster.
- Check whether sector concentration explains the grouping.
- Review how preprocessing and feature selection affect results.
- Use cluster membership for exploration, not as a final company judgment.

Model awareness

KMeans is sensitive to scaling, feature selection and the chosen number of clusters. Consequently, cluster labels are descriptive and should be interpreted together with the underlying metrics.

12 — Reports, Exports & API

Major project artifacts

Artifact	Purpose
output/screener_output.xlsx	Screening results by preset
output/peer_comparison.xlsx	Peer comparison results
output/valuation_summary.xlsx	Valuation summary
output/valuation_flags.csv	Valuation observations
output/cluster_labels.csv	Cluster assignments
output/pros_cons_generated.csv	NLP-generated signals
reports/pytest_report.html	Automated QA report
docs/analyst_guide.pdf	Analyst documentation

API validation

The FastAPI application was loaded successfully during the project QA cycle, and route checks were performed for sector and peer-related functionality.

API regression checklist

- HTTP status is correct.
- Response structure contains expected fields.
- Unknown inputs are handled safely.
- Empty results are distinguishable from errors.
- Existing endpoints remain functional after code changes.

13 — Testing & Quality Assurance

Quality assurance is essential because a technically functioning dashboard can still produce misleading results if its input data, calculations or filters are incorrect.

Testing layers

Layer	Examples
Data validation	Schema, nulls, duplicates, source completeness
Database	Tables, records, relationships, integrity
Analytics	Ratio formulas, edge cases, derived KPIs
Screener	Thresholds, presets, regression cases
Peer engine	Peer assignment and percentile outputs
Valuation	Output shape, denominator handling, flags
Clustering	Label count, assignment completeness
API	Routes, response structure, invalid inputs
UI regression	Filters, tables, key controls

Recommended execution

```
pytest -v
```

For an HTML report with pytest-html installed:

```
pytest --html=reports/pytest_report.html --self-contained-html
```

Known QA observation

A previous dashboard QA cycle identified an ICR slider omission in the screener page. This should be treated as a regression-test case whenever the screener interface is modified.

14 — Analyst Operating Procedure

The following procedure provides a professional repeatable workflow for reviewing a company or analytical universe.

01	Validate the dataset Confirm source period, company count, completeness and active filters.
02	Start with fundamentals Review company identity, sector and primary financial statements.
03	Read the ratios Inspect profitability, liquidity, leverage, efficiency and valuation metrics.
04	Run the screener Use presets or custom thresholds to identify an investigation set.
05	Compare peers Check relative positioning and investigate material differences.
06	Review valuation Examine multiples and valuation flags with context.
07	Inspect cash flow Assess operating cash generation and major investing/financing movements.
08	Review NLP signals Use extracted pros/cons as prompts for source-text review.
09	Inspect clustering Use cluster profiles to discover similar analytical patterns.
10	Document evidence Export results and retain the relevant data period, filters and assumptions.

15 — Limitations, Governance & Best Practices

Known limitations

- Data quality depends on the completeness and consistency of source files.
- Different reporting periods can make direct comparison misleading.
- Accounting definitions can vary across companies.
- Market-based metrics change with price and market conditions.
- NLP outputs depend on source text and extraction quality.
- Clustering depends on feature selection, scaling and model configuration.
- Automated flags require analyst review.

Governance principles

- Preserve raw source data separately from processed outputs.
- Never overwrite raw data during cleaning.
- Document assumptions used in derived metrics.
- Keep regression tests for every important calculation.
- Record data periods in exported reports.
- Make missing data visible.
- Separate analytical evidence from conclusions.

Best-practice checklist

Check	Before finalizing an analysis
Data period	Confirmed
Company universe	Confirmed
Missing records	Reviewed
Filters	Recorded
Ratios	Cross-checked
Peer group	Appropriate
Valuation inputs	Validated
Outliers	Investigated
Exports	Saved/versioned
QA suite	Run after material code changes

Final principle: the dashboard should make analysis faster and more reproducible without hiding uncertainty.

Appendix — Quick Reference

Project directories

data/raw — source datasets

data/processed — cleaned/derived datasets

db — SQLite database

src — application and analytics code

tests — automated tests

output — generated analytical outputs

reports — QA/report artifacts

docs — documentation

Core commands

`pytest -v` — run automated tests

`pytest --html=reports/pytest_report.html --self-contained-html` — generate HTML test report

`streamlit run src/dashboard/app.py` — launch the Streamlit dashboard when using the project's configured entry point

Key project facts

Metric	Project value
Companies	92
Ratio rows	1,160
NLP signals	874
NLP pros	562
NLP cons	312
KMeans clusters	5
Largest cluster	Value Cyclicals — 45
Known ratio exceptions	ATGL, SBIN

End of Grand Analyst Guide